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Latent profile analysis of spinal deep muscle strength and physical fitness in elite Taekwondo athletes

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Abstract

In Taekwondo, core stability and physical fitness are vital for performance, with spinal deep muscles playing a key role. This study explored the relationship between spinal deep muscle strength and physical fitness in 104 adolescent Taekwondo athletes, using a cross-sectional design in a laboratory setting. Participants were classified into groups based on spinal muscle strength at various angles, measured with a Centaur 3D machine. Outcome measures included body composition (height, weight, BMI) and physical fitness (grip strength, back muscle strength, endurance, flexibility, balance, and leg strength). Latent profile analysis identified five profiles: G2 and G4 showed higher body weight and skeletal muscle mass, with G2 also having the highest grip, back, and leg strength. No significant relationship was found between spinal muscle strength and balance. The findings suggest spinal muscle strength and physical fitness do not always correlate, highlighting the need for further research to optimize training.

Keywords Latent profile analysis, Taekwondo athletes, Spinal deep muscle strength, Physical fitness

Introduction

Taekwondo requires explosive power, speed, agility, and both anaerobic and aerobic power, which are crucial elements for performance [1]. The primary technique in Taekwondo, the kick, is generated through the rotation of the torso and accounts for most of the scoring in Taekwondo competitions [2]. The muscle activation of the torso and lower body varies depending on the form of the rotation during kicking [3]. Elite athletes demonstrate high angular velocity and muscle fiber conduction speed, requiring a high level of neuromuscular adaptation [4].

Core stability is a potential factor that can enhance kicking speed and plays a central role in generating, transmitting, and controlling optimal force and movement [5, 6]. MM Panjabi [7] described core stability as the interdependent actions of three components: bone and ligament structures, muscles surrounding the spine, and the neural control system. The core muscle group, which is the origin of all forces and mobilities in the body, is surrounded by the diaphragm (acting as the roof), pelvic floor muscles (supporting the base), abdominal muscles, and muscles around the spine, which maintain intra-abdominal pressure and provide a foundation for distal movements [6].

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Moreover, core stability focuses on the local muscle system, not only providing stability but also effectively transmitting force to the arms and legs [8]. Incorporating core-strengthening exercises into general training can rapidly enhance physical fitness [9]. This chain of functions and complex factors from the body's center facilitates smooth movements, thereby affecting overall performance.

Taekwondo athletes typically exhibit characteristics such as low body fat, high anaerobic power, strength, flexibility, and balance [10, 11], which are highly correlated with trunk strength [12]. High-intensity trunk exercises performed just before competitions can positively impact performance, highlighting the importance of strengthening the muscles around the spine [6, 13].

Previous studies have analyzed the key physical fitness factors necessary for selecting elite athletes by evaluating performance and specialized fitness determinants [14, 15], aiming to elucidate the relationship between the physical fitness factors inherent to Taekwondo, performance, and fitness levels. However, researchers have argued that simple assessments of athletes' physical fitness are insufficient for evaluating their capabilities. Recently, there has been a shift towards categorizing and understanding the diverse characteristics of athletes for more precise identification.

For instance, a study on archery athletes employed Hierarchical Agglomerative Cluster Analysis (HACA) to cluster and analyze the new potential of archers, providing a novel approach to enhance their performance [16]. While HACA struggles to explain groups based on hidden variable patterns, latent profile analysis (LPA) offers a complementary approach by classifying new group types based on similar patterns among variables, using a probabilistic latent variable modeling approach [17]. However, objective measurements of physiological and biomechanical variables for analyzing new patterns remain limited. Thus, this study aimed to identify new patterns of spinal deep muscle strength in Taekwondo athletes using latent profile analysis.

Previous research systematically reviewed the physical characteristics and fitness profiles of Taekwondo athletes [18]. Numerous studies have actively explored various physical fitness traits, such as body composition, body type, strength, and anaerobic and aerobic capacities, to enhance performance [14]. Given that physical fitness is crucial for the performance of Taekwondo athletes [15], identifying new patterns based on core stability characteristics and understanding the relationship between body composition and fitness could offer a novel approach to performance enhancement.

Therefore, this study aimed to classify new groups based on the spinal deep muscle strength of adolescent male Taekwondo athletes using latent profile analysis. Additionally, this study aimed to verify the differences in body composition and physical fitness among the classified groups to determine the relationships among spinal deep muscle strength, body composition, and fitness.

Methods

Participants

Our study recruited male adolescent Taekwondo athletes registered with the Korea Taekwondo Association. Latent Profile Analysis (LPA) was used to explore and identify subgroups within the sample. To accurately determine the required sample size, we employed G*Power 3.1.9.7, with the following parameters: effect size of 0.15, alpha level of 0.05, power of 0.80, and 8 predictors. Using the Multiple Regression: Fixed model, R^2 deviation from zero function in G*Power to approximate the LPA structure, it was estimated that a minimum of 109 participants was required to achieve reliable results. To account for potential participant dropout, we recruited a total of 127 participants, ensuring adequate power for robust profile differentiation within the LPA model. After excluding 23 dropouts, 104 participants were included in the analysis. The exclusion criteria for study participants are listed in (Table 1). Prior to the study, the purpose and intent of the research were fully explained to the participants, and consent was obtained their consent from those who voluntarily wished to participate. This study was approved by the Institutional Review Board of Dongguk University (DUIRB-202212–17) and adhered to the guidelines and

Table 1 Participants inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
1. Individuals registered as athletes with the Korea Taekwondo Association 2. Male and female high school students residing in S city 3. Individuals willing to voluntarily participate in this study and have completed the consent form	1. Individuals with metabolic disorders such as hypertension and diabetes 2. Individuals aware of cardiovascular and musculoskeletal disorders 3. Individuals with cerebral vascular diseases and mental disorders such as depression 4. Individuals who have taken medications that may affect psychiatric drug trials within the past 2 months 5. Individuals deemed unsuitable for the study by the principal investigator

ethical principles of the Declaration of Helsinki. The participants' characteristics are listed in (Table 2).

Measurement

All study participants fasted for more than 8 h before measurements were conducted between 8:00 and 10:00 a.m. The specific measurement methods are as follows.

Body composition

Height (cm) and weight (kg) were measured using a digital stadiometer (BSM 370; InBody, Seoul, Korea). Body composition was measured using a body composition analyzer (InBody 620; InBody, Seoul, Korea). Participants were asked to wear light clothing after removing any metal accessories. Body mass index (BMI) was calculated using the formula: weight (kg) / height $(m)^2$.

Physical fitness (grip strength, back muscle strength, muscle endurance, flexibility, balance, and leg strength)

Muscle Strength (Grip Strength, Back Muscle Strength, Leg Strength)

(1) Grip Strength: Measured using a handgrip dynamometer (T.K.K. 5401; Takei, Tokyo, Japan). Participants adjusted the dynamometer to fit their hands and stood comfortably with their arms straight and slightly away from their bodies at a 15° angle. The dynamometer was gripped with the second joint of the fingers at a right angle, and the grip strength of the dominant hand was measured.
(2) Back Muscle Strength: Measured using a back and leg strength dynamometer (T.K.K. 5402; Takei, Tokyo, Japan). Participants stood comfortably on the machine, with knees and torso straight, holding the handle at thigh level. The handle was pulled vertically upward without bending the elbows or tilting the body backward.
(3) Leg Strength: Measured using a leg extension machine (T.K.K. 5710M; Takei, Tokyo, Japan) in conjunction with the T.K.K. 5402. Participants sat with their backs and hips against the backrest, with the pad positioned on the lower tibia, and extended their knees.

All muscle strength measurements were performed twice with a 1-min rest in between, and the higher value was recorded for the analysis.

Muscular endurance Sit-up: To assess abdominal muscle endurance, participants lay on a mat with their feet 30 cm apart, knees bent at 90°, and hands clasped behind their heads. They raised their upper bodies to touch their elbows to the knees and return to the starting position. This was repeated for 1 min, and the number of repetitions was recorded.

Flexibility Sit-and-Reach: A flexibility testing device (T.K.K. 5403, Takei, Tokyo, Japan) was used to measure the flexibility of the lower back and hamstrings. Participants sat without shoes, with their feet against the device, and extended their hands as far forward as possible. Measurements were performed twice, with the higher value recorded.

Balance One-Legged Stance with Closed Eyes: To assess balance, participants stood comfortably with their hands on their hips and one knee bent at approximately 90° while their eyes were closed. The time until the participants opened their eyes or their lifted foot touched the ground was measured. The test was performed thrice, and the highest value was recorded. To prevent injury, a researcher assisted without interfering with the measurements.

Deep spinal muscle strength Spinal deep muscle strength was measured using a Centaur 3D machine (Bio-Feedback Motor Control, Germany), a device that assess the isometric strength of the lower back muscles in 3D space. Participants stood on the central platform with their lower body and hips fixed, chin tucked, hands on the abdomen, and the transversus abdominis contracted. Measurements were taken at eight angles in the transverse plane (0°, + 45°, -45°, + 90°, -90°, + 135°, -135°, and 180°), with clockwise directions denoted as "+" and counterclockwise as "-." Measurements were taken while the body was tilted up to 90° in the sagittal plane (Fig. 1). Participants were instructed to stabilize their upper body against gravity along the body axis. Data values were zeroed before collection, and measurements were immediately stopped upon request from the participants.

Table 2 Characteristics of participants ($n = 104$)

Variables	Mean	SD
Age (years)	16.53	0.87
Height (cm)	174.35	7.38
Weight (kg)	67.52	9.04
BMI (kg/m^2)	23.30	3.89
Body fat (kg)	20.02	6.83
Skeletal muscle (kg)	30.44	5.88

Values are presented as mean standard deviation

BMI Body Mass Index

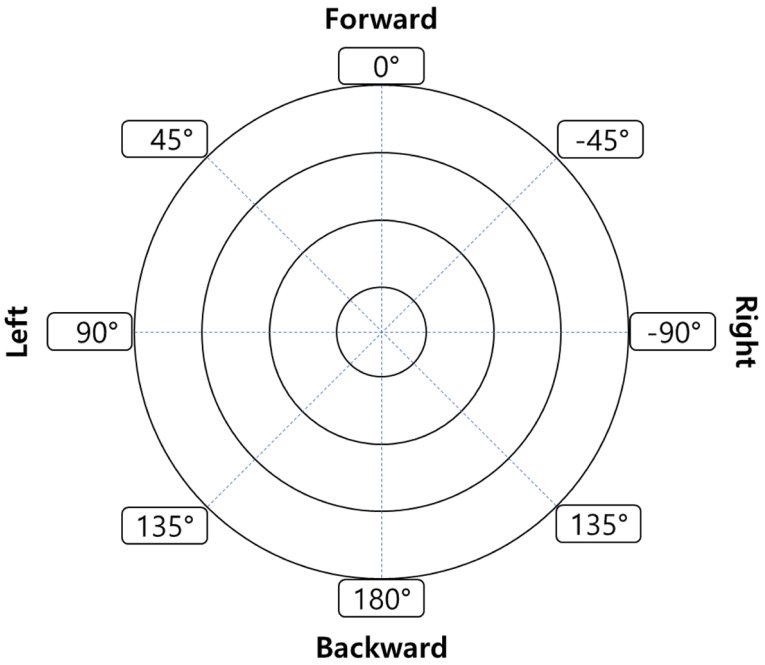


Fig. 1 Measurement positions of Centra 3d machine

Statistical analysis

To analyze differences in body composition and physical fitness based on the latent profile classification of spinal deep muscle strength in adolescent Taekwondo athletes, Latent Profile Analysis (LPA) and One-way ANOVA were conducted. Analyses were performed using Jamovi 2.3.16, with the statistical significance level set at $p < 0.05$. The specific data processing methods are as follows.

Latent Profile Analysis (LPA)

Latent Profile Analysis was conducted using the snowRMM module provided by Jamovi 2.3.16 to extract groups with common characteristics while controlling for the spinal deep muscle strength of adolescent Taekwondo athletes (0°, − 45°, 45°, − 90°, 90°, − 135°, 135°, and 180°) [19, 20]. The snowRMM module, based on the R package tidy LPA, provides LPA results [21, 22], enabling the extraction of groups that commonly or heterogeneously possess spinal deep muscle strength. Indices used to determine the number of latent profiles through Latent Profile Analysis included the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), sample-size-adjusted Bayesian information criterion (SABIC), and entropy values. AIC and BIC consider both explanatory power and parsimony by accounting for the number of estimated parameters [19] and indicate the fit between the latent profile model and the data [23], with smaller values indicating better fit [19, 20]. Given that the AIC and BIC are influenced by sample size, the SABIC, which adjusts for sample size, was also considered. Entropy indicates the classification accuracy of the estimated latent profiles [19], with values closer to 1 indicating a clearer classification [20]. Therefore, this study determined the number of latent profiles by considering the AIC, BIC, and SABIC, which focus on explanatory power and parsimony, and entropy, which indicates classification accuracy [19], while also considering the appropriateness of the classification ratio [23].

One-way ANOVA

One-way analysis of variance (ANOVA) was applied to analyze the characteristics of the latent profiles derived from LPA and the mean differences in body composition and physical fitness among the classified latent profile groups. The independent variables were the latent profiles classified by LPA, and the dependent variables were body composition (weight, body fat, and skeletal muscle mass) and physical fitness (grip strength, back muscle strength, muscular endurance, flexibility, balance, and leg strength). Post hoc comparisons were conducted using the Games-Howell method.

Results

Classification of latent profiles of spinal deep muscle strength

To classify the latent profiles of spinal deep muscle strength in adolescent Taekwondo athletes, a latent profile model was established, incorporating measurements

Table 3 Fit statistics of the latent 1 to 6 profile ($n = 104$)

Fit statistics	Profiles				
	2 Class	3 Class	4 Class	5 Class	6 Class
AIC	5496	5461	5364	5334	5337
BIC	5636	5625	5551	5546	5572
SABIC	5469	5430	5327	5293	5291
Entropy	.9901	.941	.957	.964	.962

AIC Akaike Information Criterion, *BIC* Bayesian Information Criterion, *SABIC* Sample-size Adjusted BIC

at 0°, − 45°, 45°, − 90°, 90°, − 135°, 135°, and 180°. A range of latent profiles, from one to six, was exploratorily extracted to assess model fit, and the classification results with the most appropriate explanatory power were selected. After conducting LPA, it was observed that both the AIC and BIC values gradually decreased with the extraction of up to five extracted group profiles. However, these values increased when six latent profiles were extracted. Entropy was the highest when five latent profiles were extracted. Therefore, this study determined that the model with five latent profiles had the best fit indices and that the classification ratios were appropriate as follows: Group 1, 25 participants (24.0%); Group 2, 10 participants (9.6%); Group 3, 28 participants (26.9%); Group 4, 16 participants (15.4%); and Group 5, 25 participants (24.0%). Thus, the model with five latent profiles was selected as the final model (Table 3).

Classification of latent profiles of spinal deep muscle strength (ANOVA)

Given the extraction of five latent profiles for spinal deep muscle strength in adolescent Taekwondo athletes, the characteristics of the classified groups were visualized as shown in (Fig. 2), and the analysis results are presented in (Table 4). The analysis revealed statistically significant differences in all variables of spinal deep muscle strength according to the latent profiles ($p < 0.001$). Based on post hoc comparisons, the main characteristics of each group were as follows:

First, G1, G3, and G5 exhibited similar patterns of spinal deep muscle strength, but in the following order: G1 > G3 > G5. Specifically, at 0°, the three groups were identical, but from -45° to 180°, G1 displayed the highest levels, G3 showed moderate levels, and G5 showed the lowest levels. Thus, G1, G3, and G5 can be considered groups with similar patterns of spinal deep muscle strength, but with relative differences in the amount of strength at each angle.

G2 exhibited generally high levels of spinal deep muscle strength, with particularly high values at -135° and 180°, and the second highest value at 135°. In other words, G2 showed a similar overall pattern of spinal deep muscle strength to G1 but had the highest strength at -135° and 180° and lower strength at 135° compared to G1. Finally, G4 exhibited the lowest overall spinal deep muscle strength, with higher values at 90° and 135° than G5. Therefore, G4 was characterized as having the lowest overall spinal deep muscle strength, but higher strength than G5 at 90° and 135°. The groups were named as follows: G1, high spinal deep muscle strength with good left–right balance group, G2 as the high spinal deep muscle strength with better left (-) strength group, G3 as the moderate spinal deep muscle strength with good left–right balance group, G4 as the low spinal deep muscle strength with better right (+) strength; and G5, low spinal deep muscle strength with better left (-) strength.

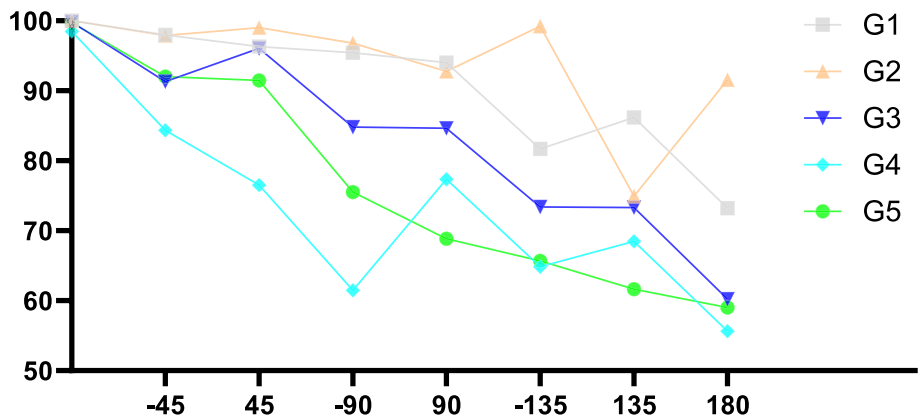


Fig. 2 Spin stabilizing muscle categories by latent class for the overall sample. G1: high spinal deep muscle strength with good left–right balance group. G2: high spinal deep muscle strength with better left-side (-) strength group. G3 had moderate spinal deep muscle strength with good left–right balance group. G4: low spinal deep muscle strength with better right-side (+) strength group. G5: low spinal deep muscle strength with better left-side (-) strength group

Table 4 Group classification results based on the number of latent profiles classified

Variables	Group	n	M	SD	F	*p*	*post-hoc*
0°	G1	25	100.00	.00	5.696	< .001	4 < 1,2,3,5
	G2	10	100.00	.00			
	G3	28	99.82	.54			
	G4	16	98.46	2.63			
	G5	25	99.72	.60			
-45°	G1	25	97.96	2.93	8.726	< .001	4 < 3,5 < 1,2
	G2	10	97.90	4.43			
	G3	28	91.33	10.46			
	G4	16	84.36	10.92			
	G5	25	92.03	6.26			
45°	G1	25	96.32	3.05	46.885	< .001	4 < 5 < 1,2,3
	G2	10	99.04	2.02			
	G3	28	96.06	3.72			
	G4	16	76.51	7.34			
	G5	25	91.49	7.46			
-90°	G1	25	95.45	5.12	79.183	< .001	4 < 5 < 3 < 1,2
	G2	10	96.82	.82			
	G3	28	84.81	7.44			
	G4	16	61.47	10.05			
	G5	25	75.52	6.27			
90°	G1	25	94.03	4.39	37.734	< .001	5 < 4 < 3 < 1,2
	G2	10	92.76	2.31			
	G3	28	84.64	6.17			
	G4	16	77.34	14.11			
	G5	25	68.84	8.57			
-135°	G1	25	81.68	8.81	34.822	< .001	4,5 < 3 < 1 < 2
	G2	10	99.24	1.60			
	G3	28	73.39	9.35			
	G4	16	64.87	12.73			
	G5	25	65.68	6.62			
135°	G1	25	86.20	4.44	70.075	< .001	5 < 4 < 2,3 < 1
	G2	10	74.94	3.01			
	G3	28	73.32	6.50			
	G4	16	68.47	5.85			
	G5	25	61.64	5.00			
180°	G1	25	73.21	7.29	70.694	< .001	4 < 3 < 1 < 2
	G2	10	91.56	5.68			
	G3	28	60.23	5.43			
	G4	16	55.64	6.85			
	G5	25	59.00	6.51			

Values are presented as mean standard deviation. One-way ANOVA with Games-Howell post hoc test was performed

Differences in body composition according to latent profile types

To determine whether the characteristics of the derived groups were related to body composition, mean differences in weight, body fat, and skeletal muscle mass according to the latent profiles were analyzed (Table 5). The analysis results revealed significant differences in weight: G1 = 69.52 ± 11.23 kg, G2 = 76.26 ± 7.53 kg, G3 = 66.56 ± 8.15 kg, G4 = 71.85 ± 6.41 kg, G5 = 66.46 ± 4.74 kg ($F = 3.778$, $p < 0.01$). Significant differences were also found in skeletal muscle mass: G1 = 31.33 ± 6.27 kg, G2 = 34.29 ± 3.68 kg, G3 = 30.59 ± 6.21 kg, G4 = 34.00 ± 3.57 kg, G5 = 29.15 ± 4.77 kg ($F = 2.922$, $p < 0.05$). However, no statistically significant differences were observed in body fat.

Table 5 Group classification results based on the number of latent profiles classified

Variables	Group	n	M	SD	F	*p*	*post-hoc*
Weight (kg)	G1	25	69.52	11.23	3.778	.007	1,3,5 < 2,4
	G2	10	76.26	7.53			
	G3	28	66.56	8.15			
	G4	16	71.85	6.41			
	G5	25	66.46	4.74			
Body fat (kg)	G1	25	20.42	5.17	.722	.579	-
	G2	10	18.00	4.17			
	G3	28	19.27	7.20			
	G4	16	17.72	4.49			
	G5	25	17.97	7.62			
Skeletal muscle (kg)	G1	25	31.33	6.27	2.922	.025	5 < 2,4
	G2	10	34.29	3.68			
	G3	28	30.59	6.21			
	G4	16	34.00	3.57			
	G5	25	29.15	4.77			

Values are presented as mean standard deviation. One-way ANOVA with Games-Howell post hoc test was performed.

Post-hoc analysis showed that weight was relatively higher in G2 and G4 than in the other groups and that skeletal muscle mass was higher in G2 and G4 than in G5.

Differences in physical fitness according to latent profile types

The analysis of mean differences in physical fitness according to latent profiles of spinal deep muscle strength in adolescent Taekwondo athletes (Table 6) showed significant differences in grip strength: G1 = 37.66 ± 8.93 kg, G2 = 40.23 ± 2.23 kg, G3 = 37.33 ± 8.65 kg, G4 = 40.61 ± 4.36 kg, G5 = 33.29 ± 6.58 kg (F = 3.104, $p < 0.05$). Significant differences were also found in back muscle strength: G1 = 94.28 ± 20.78 kg, G2 = 118.32 ± 1.16 kg, G3 = 86.80 ± 27.88 kg, G4 = 103.66 ± 18.15 kg, G5 = 92.32 ± 17.55 kg (F = 4.917, $p < 0.01$). Leg strength showed significant differences: G1 = 44.98 ± 10.23 kg, G2 = 54.11 ± 7.17 kg, G3 = 40.81 ± 11.77 kg, G4 = 42.63 ± 6.59 kg, G5 = 40.84 ± 8.27 kg (F = 4.316, $p < 0.01$). Muscular endurance also showed significant differences: G1 = 47.60 ± 7.90 reps, G2 = 52.80 ± 5.25 reps, G3 = 49.96 ± 6.74 reps, G4 = 44.63 ± 15.79 reps, G5 = 39.76 ± 10.93 reps (F = 5.050, $p < 0.01$). Flexibility showed significant differences as well: G1 = 21.52 ± 9.27 cm, G2 = 16.82 ± 8.42 cm, G3 = 25.45 ± 6.85 cm, G4 = 20.28 ± 8.56 cm, G5 = 21.72 ± 6.43 cm (F = 2.683, $p < 0.05$). However, no significant differences were observed in balance. Post hoc comparisons indicated that grip strength was higher in G2 and G4 than in G5, and back muscle strength was higher in G2 than in G3 and G4. Muscular endurance was higher in G2 than in G5, whereas flexibility was the highest in G3. Additionally, leg strength was highest in G2 compared to other groups.

Discussion

The aim of this study was to classify new groups of adolescent male Taekwondo athletes based on their spinal deep muscle strength using latent profile analysis, and to verify differences in body composition and physical fitness among the classified groups. Five profiles were extracted. The classified groups showed the following characteristics: G1, high spinal deep muscle strength with good left–right balance; G2, high spinal deep muscle strength with better left (-) strength; G3, moderate spinal deep muscle strength with good left–right balance; G4. low spinal deep muscle strength with better right (+) strength; and G5, low spinal deep muscle strength with better left (-) strength. This study demonstrated that adolescent male Taekwondo athletes exhibit different characteristics based on spinal deep muscle strength. This attribute serves as a key indicator of core stability and reflects differences in body composition and physical fitness variables. The results are discussed below.

First, groups G1, G3, and G5 showed consistent decreases in spinal deep muscle strength from 0° to 180°, with G1 > G3 > G5 in terms of strength. G1 exhibited high strength, G3 showed moderate strength, and G5 showed low strength, respectively. G2 and G4 exhibited different characteristics depending on the angle. G2 generally exhibited high strength, with particularly high values on the right (+) side and at 180°. In contrast, G4 generally demonstrated lower strength, surpassing G5 only at 90° and 135°. A notable finding is that even among adolescent male Taekwondo athletes, spinal deep muscle strength can be categorized into various types. This indicates that

Table 6 Group classification results based on the number of latent profiles classified

Variables	Group	n	M	SD	F	*p*	*post-hoc*
Handgrip strength (kg)	G1	25	37.66	8.93	3.104	.019	5 < 2,4
	G2	10	40.23	2.23			
	G3	28	37.33	8.65			
	G4	16	40.61	4.36			
	G5	25	33.29	6.58			
Back strength (kg)	G1	25	94.28	20.78	4.917	.001	3 < 4 < 2
	G2	10	118.32	1.16			
	G3	28	86.80	27.88			
	G4	16	103.66	18.15			
	G5	25	92.32	17.55			
Leg strength (kg)	G1	25	44.98	10.23	4.316	.003	3,5,4,1 < 2
	G2	10	54.11	7.17			
	G3	28	40.81	11.77			
	G4	16	42.63	6.59			
	G5	25	40.84	8.27			
Flexibility (cm)	G1	25	21.52	9.27	2.683	.036	2 < 3
	G2	10	16.82	8.42			
	G3	28	25.45	6.85			
	G4	16	20.28	8.56			
	G5	25	21.72	6.43			
Balance (sec)	G1	25	22.20	7.90	1.372	.249	-
	G2	10	21.40	13.95			
	G3	28	19.43	9.49			
	G4	16	18.25	9.98			
	G5	25	24.04	6.52			
Abdominal muscle endurance (rep)	G1	25	47.60	7.90	5.050	.001	5 < 2
	G2	10	52.80	5.25			
	G3	28	49.96	6.74			
	G4	16	44.63	15.79			
	G5	25	39.76	10.93			

Values are presented as mean standard deviation. One-way ANOVA with Games-Howell post hoc test was performed

despite the sports-specific nature of Taekwondo, athletes exhibit different patterns of spinal deep muscle strength.

For Taekwondo athletes, the function of the spinal deep muscles in maintaining posture stability varies by age group and is more critical for athletes in their 20 s compared to junior athletes [24]. However, adolescence is a transitional period until adulthood, and it is crucial not to overlook the stabilization muscles during this stage. These characteristics likely reflect differences in foot use, kicking techniques, and bodily functions among Taekwondo athletes. Through latent profile analysis and consideration of body composition information, we found that the profiles of G2 and G4 were strongly correlated with weight and skeletal muscle mass. Further analysis of how different patterns of spinal deep muscle strength influenced physical fitness showed that G2 exhibited relatively higher grip strength, back muscle strength, muscular endurance, and leg strength than the other groups. However, flexibility was the lowest in this group. Moreover, excluding G2, G4 displayed higher grip strength and back muscle strength than the other groups. Interestingly, despite having relatively lower spinal deep muscle strength, G4 exhibited higher grip strength and back muscle strength than the other groups. This indicates that the type of spinal deep muscle strength profile affects body composition and physical fitness factors, although it does not affect balance. The distinct characteristics of each latent profile, such as G2's increased spinal muscle strength at specific angles, provide valuable insights for designing targeted training interventions tailored to unique muscle strength profiles [25]. The observed consistent decline in spinal deep muscle strength from 0° to 180° in certain groups may be due to variations in

muscle fiber recruitment at different joint angles [26, 27]. This suggests that different groups may exhibit linear or non-linear muscle behavior based on factors such as their level of muscle development, resistance to muscle fatigue, or specific movement patterns associated with Taekwondo. Further research, including electromyography (EMG) analysis, is recommended to clarify the variations in muscle fiber recruitment across different joint angles and to determine their influence on these distinct muscle behavior patterns.

In university football players, the cross-sectional area and contraction thickness of the multifidus muscle are positively correlated with body weight, body fat mass, and muscle mass [28]. Similarly, in ice hockey players, the echo intensity (EI) of the multifidus muscle showed a strong correlation with body fat percentage, body fat mass, and lean body mass [29]. Our findings align with previous research, demonstrating that groups with greater muscle strength are more strongly associated with body weight and skeletal muscle mass. However, our study did not observe a significant correlation between muscle strength and body fat. This may be attributed to the relatively low body fat levels typically found in Taekwondo athletes, which could obscure significant differences that might otherwise be detected [30]. Furthermore, unlike prior studies that employed ultrasound echo intensity to assess fatty infiltration in muscle tissue, our study did not utilize this technique, potentially limiting our ability to identify correlations between body fat and spinal muscle composition.

Taekwondo requires anaerobic power, strength, flexibility, and explosiveness [10, 11], and improving core function through core exercises can enhance back muscle strength, lower body strength, and grip strength, which are key muscles of the body [31, 32]. Our study also supports previous findings, as group G2 exhibited high grip strength, back muscle strength, muscular endurance, and leg strength. A notable finding in our study was that despite lower spinal deep muscle strength, G4 exhibited higher grip strength and muscular endurance than the other groups. This may be because grip strength, a major tool for predicting adolescent strength, is heavily influenced by body weight [33]. G3 exhibited the highest flexibility. Despite having moderate spinal deep muscle strength, lower weight, and back muscle strength, this group showed the highest flexibility. This result is significant given that flexibility is a critical factor for sports requiring high levels of flexibility, highlighting variations within the average level. However, a meta-analysis of the relationship between trunk strength and physical fitness indicated a higher correlation among recreational participants than elite athletes [12]. These results suggest the need for more comprehensive studies on the relationship between the various functions of spinal deep muscles and physical fitness in athletes.

No significant differences in balance were found based on the latent profile analysis, consistent with previous studies. For instance, core strength in lacrosse players was not correlated with balance [34], and no correlation was found between core muscular endurance and balance in university lacrosse and soccer players [35]. These results contradict the common findings that core stabilization exercises enhance strength, endurance, and balance [36, 37]. This suggests that other factors, beyond trunk strength or muscular endurance which were not assessed in this study may play a critical role in influencing balance.

However, research on the relationship between core muscles and balance remains limited, indicating the need for further studies to clarify this relationship. The lack of a relationship between balance and deep spinal muscle strength is a significant point of consideration. Taekwondo is characterized by dynamic movements, requiring fast and high-rotating kicks, and diverse movement structures [15]. Considering the varying activation levels of core and leg muscles during different kicks, tailored training addressing these aspects is necessary[4].

This study has several limitations. First, it focused exclusively on adolescent male Taekwondo athletes, which means the results may differ based on sex and age. Therefore, future research should include female athletes to investigate sex-based differences in spinal muscle strength and fitness. Additionally, grouping similar content could help improve the clarity of the study. Second, the analysis was limited to latent profile analysis based on spinal deep muscle strength, necessitating further research on other physiological variables in Taekwondo athletes. Third, comparisons of dominant and non-dominant limb strengths were not included. Nevertheless, this study is the first to reclassify adolescent male taekwondo athletes based on spinal deep muscle strength using latent profile analysis. Additionally, it provides new insights into body composition and physical fitness that differ from those of previous studies. Building on the findings of this study, future research could further investigate ways to enhance core muscle strength and overall physical fitness in athletic performance. One such potential area of exploration involves the use of functional sports drinks, such as those containing B240 probiotics, which could enhance athletic recovery and performance by supporting immune function, reducing inflammation, and improving gut health [38]. These benefits might indirectly influence muscle recovery and overall physical fitness, making such drinks a complementary option for elite athletes aiming to optimize their training and

recovery processes. Further research should explore their direct impact on specific parameters like muscle strength and endurance.

Conclusion

In this study classified adolescent male taekwondo athletes into five groups based on their spinal deep muscle strength using latent profile analysis. Groups G2 and G4, which had different spinal deep muscle strength characteristics, had greater weights and skeletal muscle masses than the other groups. Additionally, G2 and G4 exhibited higher grip strength and back muscle strength, indicating that weight and skeletal muscle mass might have a greater influence on these metrics more than deep spinal muscle strength. Finally, no relationship was found between spinal deep muscle strength and balance. These findings suggest that lower spinal deep muscle strength does not necessarily correlate with lower physical fitness in adolescent Taekwondo athletes, and differences in left–right spinal deep muscle strength do not seem to affect balance. This emphasizes the need for individualized training to enhance Taekwondo athlete performance.

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Authors' contributions
J.-H.L. and M-S.H. designed the study, with input regarding feasibility from T.L. All authors were involved in the data collection (J.-H.L. and T.L. [recruitment and data collection of players]). T.L. performed the statistical analysis. J.-H.L. and T.L. interpreted the data and J.-H.L. M.L. and M.-S.H. wrote the manuscript, which was revised by M.L. and M.-S.H. All authors read and approved the final manuscript.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study involved 104 male adolescent taekwondo athletes registered with the Korea Taekwondo Association. Written informed consent was obtained from all participants prior to their inclusion in the study. Before providing their written informed consent, all players were fully informed about the study's purpose, protocol, associated benefits, and potential risks. This study was approved by the Institutional Review Board of Dongguk University (DUIRB-202212–17) and adhered to the guidelines and ethical principles of the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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